DO NOW — RECALL

Lesson 073 · Trace Tables and Boundary Testing

Name: _____ Class: _____ Date: _____

1. What does CountPasses count?

2. What is a trace table for?

3. What is a boundary condition?

4. Stack Push 3, Push 7, then Pop. What is popped and what remains?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Array values greater than or equal to40.

2. Model answer

Showing changing variables/decisions.

3. Model answer

A limit at which program behaviour changes.

4. Model answer

Popped 7; the stack contains 3.

By the end, you can…

• A trace table records variable values after each meaningful execution step: i, a[i], condition and result.

• A boundary test probes just below, at and above a threshold. Normal, erroneous and exceptional tests reveal different faults.

• Test early success, late success, absent target, and empty arrays. For this function, −1 distinguishes not found from index0.

Vocabulary — know these terms

trace table — State changes in algorithm execution.

sentinel — Special value indicating a condition.

boundary test — Values at or near limit.

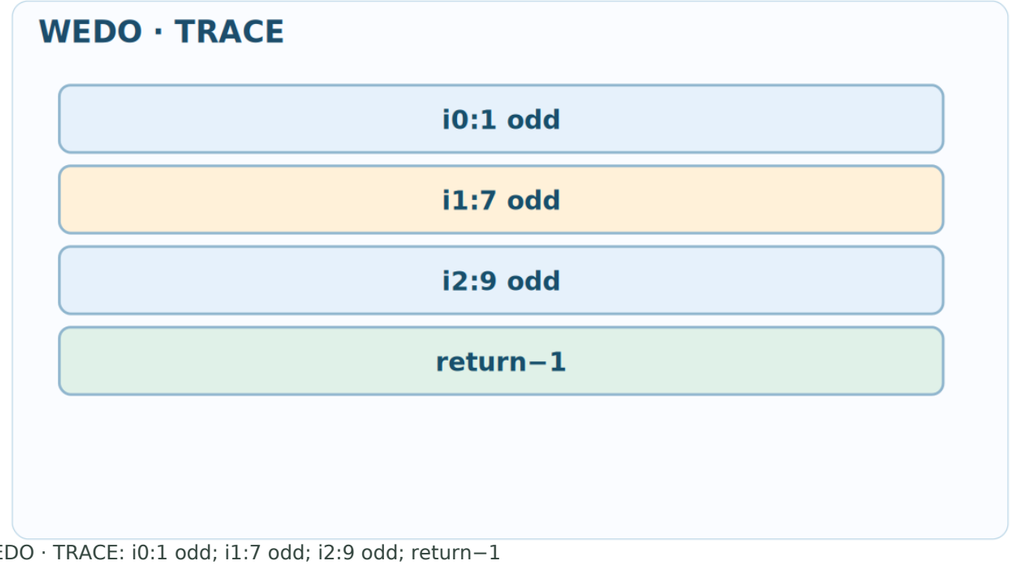
exceptional test — Unusual or error-causing case.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace first even index in [1,7,9], including return when absent.

No candidate satisfies value%2==0, so the not-found sentinel is returned.



WEDO · TRACE: i0:1 odd; i1:7 odd; i2:9 odd; return−1

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Explain
No candidate satisfies value%2==0, so the not-found sentinel is returned.

NEW LEARNING

Read, annotate and explain

Observe

A trace table records variable values after each meaningful execution step: i, a[i], condition and result.

Explain

A boundary test probes just below, at and above a threshold. Normal, erroneous and exceptional tests reveal different faults.

Evaluate

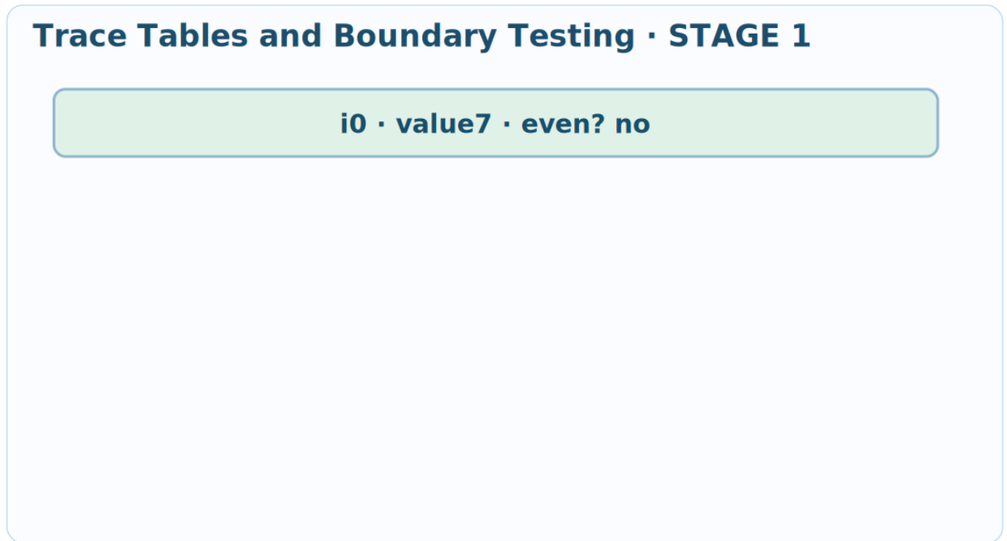
Test early success, late success, absent target, and empty arrays. For this function, −1 distinguishes not found from index0.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Trace Tables and Boundary Testing · STAGE 1: i0 · value7 · even? no

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Returning0 is a safe 'not found' value for zero-indexed arrays.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

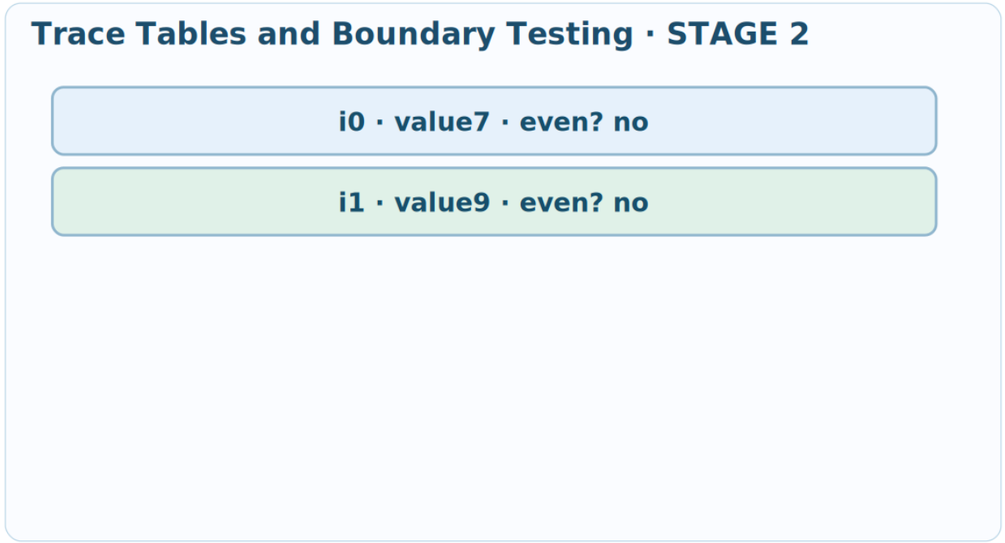
Question 3

A program incorrectly returns0 when there is no even value. Explain the ambiguity and propose a safer fix, with tests. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

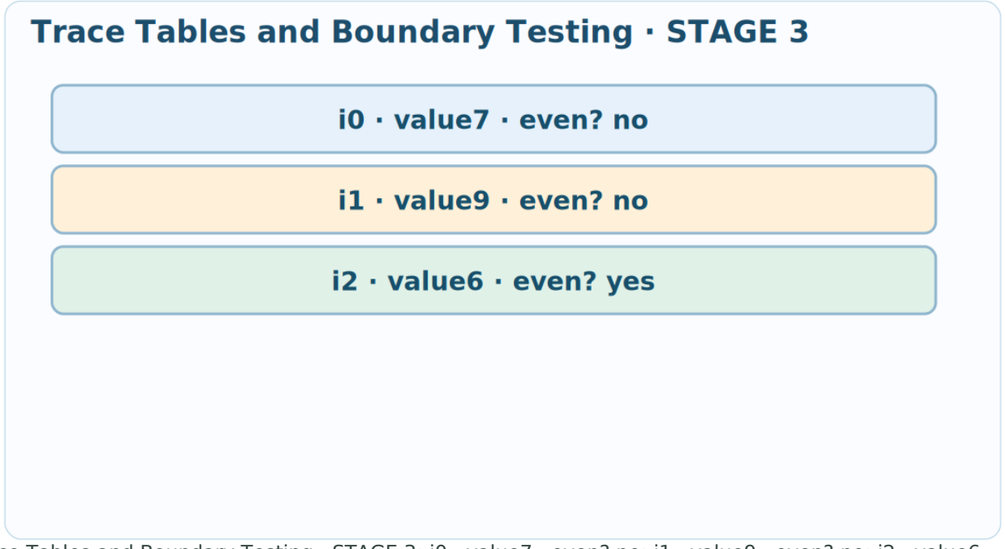


Trace Tables and Boundary Testing · STAGE 2: i0 · value7 · even? no; i1 · value9 · even? no

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Trace Tables and Boundary Testing · STAGE 3: i0 · value7 · even? no; i1 · value9 · even? no; i2 · value6 · even? yes

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace FindFirstEven([7,9,6,4]) with index and parity. [6]

Question 2

Identify expected outputs for [], [2,3], [3,5,8], [1,7,9]. [4]

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

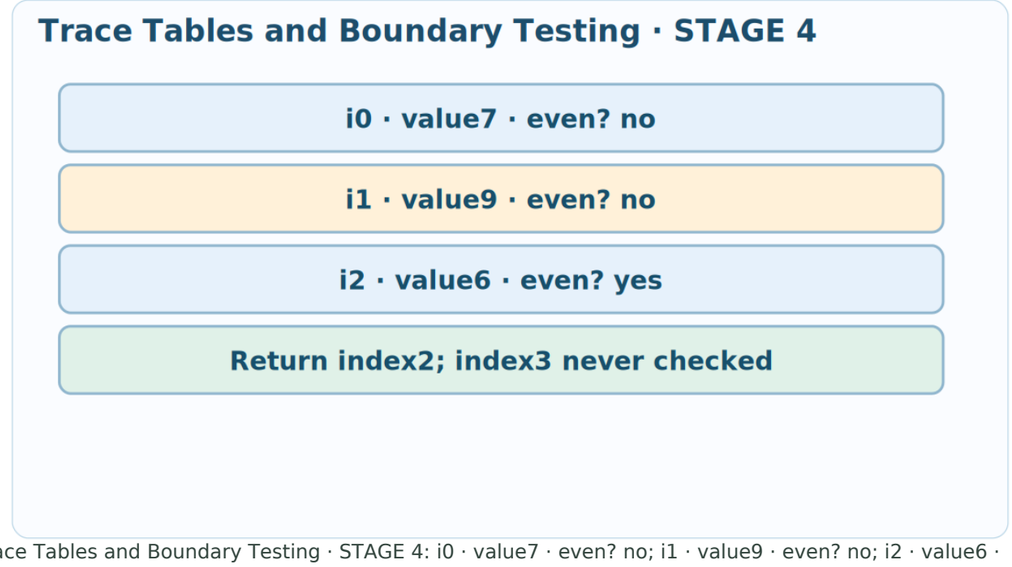
Guided 1 — TRY

Trace first even index in [1,7,9], including return when absent.
Source code / input:
i0 1 odd; i1 7 odd; i2 9 odd; loop ends, returns−1.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Trace Tables and Boundary Testing · STAGE 4: i0 · value7 · even? no; i1 · value9 · even? no; i2 · value6 · even? yes; Return index2; index3 never checked

I DO — WORKED EXAMPLE

Follow the worked method

Trace first even index in [3,5,8,10].

Explain

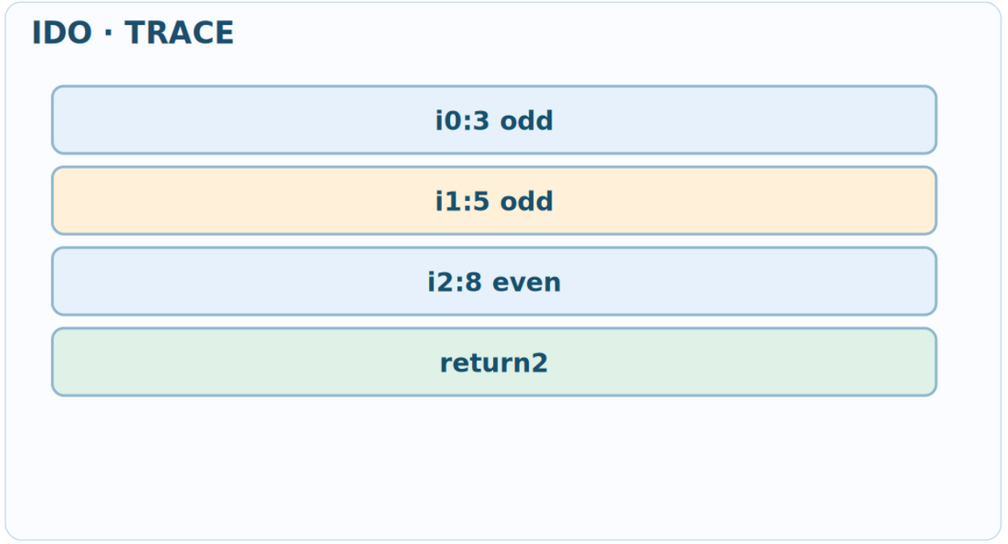
The function exits at first even occurrence and does not examine10.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace first even index in [3,5,8,10].

The function exits at first even occurrence and does not examine10.



IDO · TRACE: i0:3 odd; i1:5 odd; i2:8 even; return2